EV Wake-Up via Control Pilot — functional description

Some electric vehicles enter a deep sleep when a charging session is paused (IEC 61851 state B: vehicle connected, not charging — for example during load balancing, when the building temporarily needs the available current). Once asleep, these vehicles do not respond when the charger re-advertises current on the Control Pilot (CP), so charging never resumes on its own.

To recover from this, the charger can perform a **CP disconnect**: a brief, deliberate interruption or alteration of the pilot signal that makes the vehicle believe the cable was unplugged and re-plugged, causing it to re-run its connection sequence and wake up. Three variants exist; which one is used is a per-installation configuration choice.

Variant 1 — physical interruption

The CP conductor is galvanically interrupted for a few seconds (default 3 s) and then restored. From the vehicle's perspective this is indistinguishable from a genuine unplug/replug of the connector, so the vehicle restarts its plug-in detection.

This variant requires the optional **CP disconnect relay board**, available from Edge Technologies (support@edgetech.eu).

Variant 2 — simulated charger absence ("EV40")

The PWM on the pilot is suspended and the CP is held at a **static −12 V** — the level a vehicle interprets as "charging station not available" (state F) — for a configurable period (default 3 s). The previous pilot signal is then restored. The vehicle experiences the charger briefly disappearing and re-initiates the session when it returns.

Variant 3 — simulated unplug/replug sequence ("343S")

A fixed three-step pilot pattern of about 10 seconds total, for vehicles that need a more complete unplug signature:

1. **+12 V static** for 3 s — the idle level a charger presents with no vehicle attached, with no PWM present;
2. **−12 V static** for 4 s — "charging station not available" (state F);
3. **+12 V static** for 3 s;
4. normal pilot signalling resumes.

When it is performed

A CP disconnect is only ever performed while the vehicle is in **state B** (connected, not charging). It can be triggered two ways:

- **Automatically**: each time the session *enters* state B — at plug-in, or when charging pauses (state C → B) — a timer starts (default 30 s). If the vehicle is still in state B when it expires, one disconnect is performed. This is one-shot: it does not repeat while the vehicle remains in state B, and it is re-armed only by the next entry into state B (e.g. the next charging pause).
- **On remote command**: the back office can request a disconnect at any moment; the request is only executed while in state B and never while a disconnect is already in progress.

Both the automatic behaviour and the choice of variant are configuration options and are disabled by default.

Configuration via Modbus

Register addresses and names below follow the published register map (*ModBusRegisters for Edge Technologies' EV Charger Controller (SmartEVSE)*, v2.10.17). Communication: Modbus RTU over RS485, default address 0x01 @ 9600 8N1. Note: while the device is still at default address 0x01 it only responds to Modbus after a state change has occurred (e.g. pressing the button on the board).

1. Select the mechanism — `Settings` register 0x0204 (holding, fc 3/6)

Read the current value first, change only the bits below, and write it back — the same register also holds the phase, lock and DCL configuration.

Bit	Mask	Name	Effect
3	0x0008	`MISUSE_LOCK_PORT_AS_CP_DISCONNECT`	**Variant 1** (physical interruption): the cable-lock port drives the CP-disconnect relay instead. The lock registers (0x0010, 0x0015, 0x0016) then return 0xFFFF.
5	0x0020	`EV40_CP_DISCONNECT`	Enables the **software** disconnect (variant 2 or 3, selected below).
4	0x0010	`AUTO_CP_DISCONNECT`	Enables the **automatic** trigger (timer after entering state B). Without it, only remote-commanded disconnects occur.

2. Choose the software variant — `CPDisconnectType`, high byte of 0x020A

Register 0x020A packs two bytes: low = `CPType`, high = `CPDisconnectType`. Read-modify-write so the low byte is preserved. High byte **0 = EV40** (variant 2, the default), **1 = extended EV40** (variant 3, "343S").

3. Timing — 0x0208 and 0x0209

- `CPDisconnectTime` (0x0208): duration in ms of the disconnect for variant 1 and variant 2 (default 3000). Variant 3 uses its fixed 3 s / 4 s / 3 s pattern.
- `TimeBeforeCPDisconnect` (0x0209): the automatic-trigger wait in ms after entering state B (default 30 000).

4. Persist the settings

Register writes take effect immediately but are lost at reboot until saved:

1. Write the magic code (default 0x1234) to `MagicResetCode` (0x0202, fc 6).
2. In the **immediately following transaction**, read `SaveEEPROM` (input register 0x000A, fc 4). It returns 0x0000 on success, 0xFFFF if the unlock was missed — the unlock is valid for that next transaction only.

5. Triggering a disconnect remotely

Read input register `Disconnect CP` (0x0017, fc 4) at any time. It returns 1 when the disconnect is performed and 0xFFFF when the charger is not configured for CP disconnect (neither bit 3 nor bit 5 set). The request is only executed while the vehicle is in state B and never while a disconnect is already in progress.

Example: enable automatic 343S wake-up

1. Read 0x0204, OR in 0x0030 (bits 4+5), write it back.
2. Read 0x020A, set the high byte to 1 (keep the low byte), write it back.
3. Optionally adjust 0x0209 (wait before disconnect, ms).
4. Write 0x1234 to 0x0202, then read 0x000A → expect 0x0000.